

The Program in Action

When you run *caesarhacker.py*, the output will look like this:

```
Caesar Cipher Hacker, by Al Sweigart al@inventwithpython.com
Enter the encrypted Caesar cipher message to hack.
> QIIX QI FC XLI VSWI FYWLIW XSRMKLX.
Key #0: QIIX QI FC XLI VSWI FYWLIW XSRMKLX.
Key #1: PHHW PH EB WKH URVH EXVKHV WRQLJKW.
Key #2: OGGV OG DA VJG TQUG DWUJGU VQPKIJV.
Key #3: NFFU NF CZ UIF SPTF CVTIFT UPOJHIU.
Key #4: MEET ME BY THE ROSE BUSHES TONIGHT.
Key #5: LDDS LD AX SGD QNRD ATRGDR SNMHFGS.
Key #6: KCCR KC ZW RFC PMQC ZSQFCQ RMLGEFR.
--snip--
```

How It Works

Note that lines 20 to 36 in this program are nearly identical to lines 55 to 78 in the Caesar cipher program. The hacking program implements the same decryption code, except that it does so in a for loop, which runs the code for every possible key.

Unfortunately, the hacking program isn't sophisticated enough to identify when it has found the correct key. It relies on a human to read the output and identify which decryption produced the original English (or whichever written language was encrypted). Chapter 11 of the book *Cracking Codes with Python* (No Starch Press, 2018) details how you can write Python code to detect English messages.

```
1. """Caesar Cipher Hacker, by Al Sweigart al@inventwithpython.com
2. This program hacks messages encrypted with the Caesar cipher by doing
3. a brute force attack against every possible key.
4. More info at:
5. https://en.wikipedia.org/wiki/Caesar_cipher#Breaking_the_cipher
6. View this code at https://nostarch.com/big-book-small-python-projects
7. Tags: tiny, beginner, cryptography, math"""
8.
9. print('Caesar Cipher Hacker, by Al Sweigart al@inventwithpython.com')
10.
11. # Let the user specify the message to hack:
12. print('Enter the encrypted Caesar cipher message to hack.')
13. message = input('> ')
14.
15. # Every possible symbol that can be encrypted/decrypted:
16. # (This must match the SYMBOLS used when encrypting the message.)
17. SYMBOLS = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ'
18.
19. for key in range(len(SYMBOLS)): # Loop through every possible key.
20.     translated = ''
21.
22.     # Decrypt each symbol in the message:
23.     for symbol in message:
```